Calculation of reference lines

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SOURCE: MEPC.215(63).pdf
appendix lists the ship types from IHSF used for the calculation of reference lines. Table 2 lists the IHSF ship types not used when calculating the reference lines.

Calculation of reference lines

13 To calculate the reference line, an estimated index value for each ship contained in the set of ships per ship type is calculated using the following assumptions:

.1 the carbon emission factor is constant for all engines, i.e. CF,ME = CF,AE = CF = 3.1144 g CO2/g fuel;

.2 t he specific fuel consumption for all ship types is constant for all main engines, i.e. SFCME = 190 g/kWh;

.3 PME(i) is 75% of the total installed main power (MCRME(i));

.4 the specific fuel consumption for all ship types is constant for all auxiliary engines, i.e. SFCAE = 215 g/kWh;

.5 PAE is the auxiliary power and is calculated according to paragraphs 2.5.6.1 and 2.5.6.2 of the annex to MEPC.XXX(63);

.6 no correction factors are used; and

.7 innovative mechanical energy efficiency technology, shaft motors and other innovative energy efficient technologies are all excluded from the reference line calculation, i.e. PAEeff = 0, PPTI = 0, Peff = 0.

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SOURCE: MEPC.215(63).pdf
.7 innovative mechanical energy efficiency technology, shaft motors and other innovative energy efficient technologies are all excluded from the reference line calculation, i.e. PAEeff = 0, PPTI = 0, Peff = 0.

14 The equation for calculating the estimated index value for each ship (excluding containerships see paragraph 15) is as follows:

ref
NME
i
AEiME
VCapacity
PP
ValueIndexEstimated

15 For containerships, 70 per cent of the deadweight (70% DWT) is used as capacity for calculating the estimated index value for each containership as follows:

ref
NME
i
AEiME
V
PP
ValueIndexEstimated

DWT%70
215190
1144.3 1

Calculation of reference line parameters "a" and "c"

16 For all ship types to which these guidelines apply, parameters "a" and "c" are determined from a regression analysis undertaken by plotting the calculated estimated index values against 100 per cent deadweight (100% DWT).

Documentation

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SOURCE: MEPC.215(63).pdf
16 For all ship types to which these guidelines apply, parameters "a" and "c" are determined from a regression analysis undertaken by plotting the calculated estimated index values against 100 per cent deadweight (100% DWT).

Documentation

17 For purposes of transparency, the ships used in the calculation of the reference lines should be listed with their IMO numbers and the numerator and denominator of the index formula, as given in paragraphs 14 and 15. The documentation of the aggregated figures preserves the individual data from direct access but offers sufficient information for possible later scrutiny.

APPENDIX

1 To ensure a uniform interpretation, ship types defined in regulation 2 of MARPOL Annex VI are compared to the ship types given in the IHSF database.

2 The IHSF Stat code system provides several levels of definition as follows:

.1 Highest level:

A Cargo carrying
B Work vessel
W Non-seagoing merchant ships

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SOURCE: MEPC.215(63).pdf
MARPOL Annex VI are compared to the ship types given in the IHSF database.

2 The IHSF Stat code system provides several levels of definition as follows:

.1 Highest level:

A Cargo carrying
B Work vessel
W Non-seagoing merchant ships
X Non-merchant
Y Non-propelled
Z Non-ship structures

For the purpose of the EEDI, only group " A cargo carrying" needs to be considered. A graphical representation of this is given below.

.2 The next level comprises:

A1 Tankers
A2 Bulk carriers
A3 Dry cargo/passenger

There are further differentiations until level five, e.g. "A31A2GX General Cargo Ship", and each category is described.

The complete list is attached.

3 The ship types from the IHSF Stat code 5 (Statcode5v1075) used for the calculation of reference lines for the following ship types: bulk carrier, gas carrier, tanker, containership, general cargo ship, refrigerated cargo carrier and combination carrier, are set out in Table 1.

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SOURCE: MEPC.215(63).pdf
reference lines for the following ship types: bulk carrier, gas carrier, tanker, containership, general cargo ship, refrigerated cargo carrier and combination carrier, are set out in Table 1.
The IHSF database ship types, not used in the calculation of reference lines for the specific ship types, are set out in Table 2, e.g. ships built for sailing on the Great Lakes and landing craft.

Table 1: Ship types from IHSF used for the calculation of